

Highlights

Transport validation of a wave-coupled lagoon model: interacting effects of seagrass roughness and bed mobility

Cicero Martins Jr., Giuseppe Ciraolo, Mauro De Marchis

- Six model configurations agree on water level but differ by 30% in drifter skill
- Seagrass roughness and bed mobility change transport only when applied together
- A mobile bed corrects how far particles travel, roughness corrects where
- Only the combined configuration reproduces the observed drifter transport rate
- Surface current exceeds bed current by 1.9 to 4.5 times at sub-metre depth